

Module 3: Data Link Layer and Wireless Networks



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Data-Link Layer

- Data Link Control (DLC)
- Multiple Access Protocols (MAC)
- Link-Layer Addressing
- Ethernet Protocol
- Connecting Devices

Wireless Networks

- Wireless LANs
- Mobile IP

Hands-On

- Datalink Provider Interface
- SOCK_PACKET
- PF_PACKET

Connecting Devices

Hosts and networks rarely operate independently.

Connecting devices are used to:

- Connect hosts to form a network
- Connect networks to form an internetwork
- Extend network coverage
- Improve communication efficiency

Three Main Connecting Devices

- 1 Repeaters (Hubs)
- 2 Link-Layer Switches
- 3 Routers

Connecting Devices and Internet Layers

Device	Layer of Operation
Repeater / Hub	Physical Layer
Link-Layer Switch	Physical + Data-Link Layer
Router	Physical + Data-Link + Network Layer

More Layers = More Intelligence

Repeaters and Hubs

A repeater is a physical-layer device.

Functions:

- Receives weakened signals
- Regenerates original bit pattern
- Retimes the signal
- Sends refreshed signal forward

Used to overcome signal attenuation in long-distance communication.

Why Repeaters are Needed?

Signals lose strength while traveling through a medium.

Problems caused by attenuation:

- Weak signals
- Distorted bits
- Increased transmission errors



Solution

A repeater regenerates the original signal before it becomes unusable.

Hub: Multiport Repeater

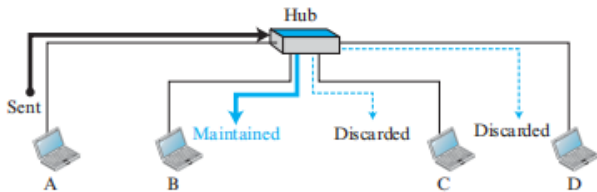
Modern Ethernet LANs use star topology.

A hub is:

- A multiport repeater
- Central connection point
- Physical-layer device



Figure 5.84 *Repeater or hub*



Working of a Hub

Suppose station A sends a frame to station B.

The hub:

- 1 Receives the signal
- 2 Regenerates the bits
- 3 Sends the frame through every port except the incoming port

All stations receive the frame.

Only station B accepts it.

Other stations discard it.

Characteristics of Hubs

- Operates only at Physical Layer
- No MAC address
- Does not inspect frames
- No filtering capability
- Broadcasts every frame
- Very simple device



Important

A hub does not know where a frame should go.

Link-Layer Switches

A Link-Layer Switch operates in:

- Physical Layer
- Data-Link Layer

Functions:

- Regenerates signals
- Examines MAC addresses
- Filters traffic
- Forwards frames intelligently



Hub vs Switch

Hub	Switch
Physical layer device	Data-link layer device
No filtering	Filtering available
Broadcasts all frames	Sends only to destination port
No MAC table	Uses MAC table
Less efficient	More efficient

Filtering in a Switch

A switch examines:

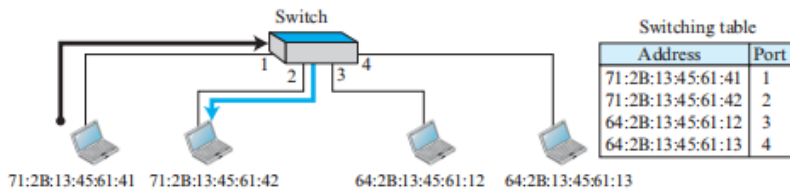
- Destination MAC Address
- Source MAC Address

Process:

- 1 Receive frame
- 2 Check destination address
- 3 Look up switching table
- 4 Send frame only through required port



Figure 5.85 *Link-Layer Switch*



Key Features of Switches

Important Facts

- Switch has filtering capability
- Uses a forwarding table
- Does not change MAC addresses
- Reduces unnecessary traffic

Transparent Switches

A transparent switch is invisible to stations.

Characteristics:

- Stations are unaware of switch presence
- No host reconfiguration required
- Switches can be added or removed easily

Defined by IEEE 802.1D.

Requirements of Transparent Switches

IEEE 802.1D specifies three requirements:

- 1 Correct forwarding of frames
- 2 Automatic learning of addresses
- 3 Prevention of loops

Learning Process in Switches

Modern switches automatically learn MAC addresses.

Switch examines:

- Destination Address → Forwarding decision
- Source Address → Learning process

Dynamic tables eliminate manual configuration.

Learning Example

Step 1

- Station A sends frame to Station D
- Switch does not know A or D
- Frame is flooded
- Learns: A is connected to Port 1

Step 2

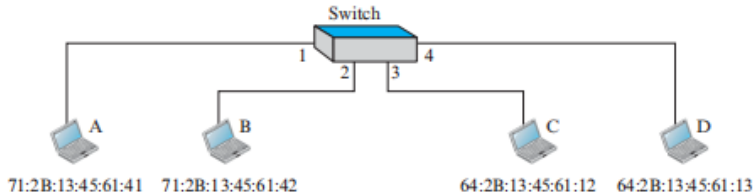
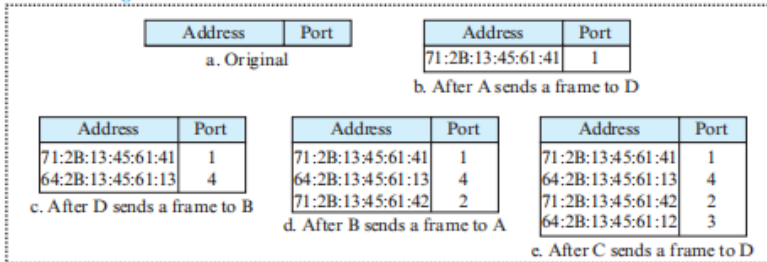
- Station D sends frame
- Switch learns D's port

Result

- Table gradually becomes complete

Figure 5.86 *Learning switch*

Gradual building of Table



Advantages of Learning Switches

- No manual MAC configuration
- Automatically adapts to network changes
- Supports plug-and-play operation
- Efficient frame forwarding

Routers

A router is a three-layer device.

Layers:

- Physical Layer
- Data-Link Layer
- Network Layer



Routers connect multiple independent networks.

Functions of a Router

As a:

Physical Layer Device

- Regenerates signals

Data-Link Layer Device

- Checks MAC addresses

Network Layer Device

- Checks IP addresses
- Makes routing decisions

Router vs Switch

Switch	Router
Operates at Layer 2	Operates at Layer 3
Uses MAC address	Uses IP address
Connects devices in same LAN	Connects different networks
Does not modify MAC addresses	Changes MAC addresses

Major Differences Between Router and Switch

Difference 1

- Router has both MAC and IP addresses on every interface.

Difference 2

- Router accepts packets only if destination MAC matches interface.

Difference 3

- Router changes source and destination MAC addresses during forwarding.

Why Routers Change MAC Addresses?

MAC addresses have only local significance.

At every hop:

- 1 Old frame removed
- 2 New frame created
- 3 New source MAC inserted
- 4 New destination MAC inserted

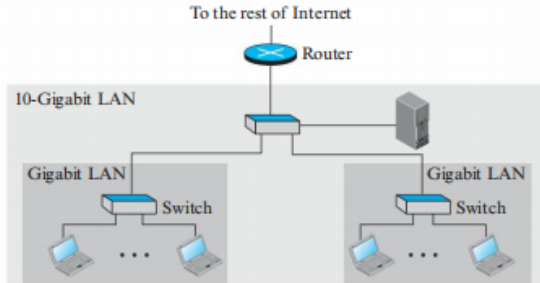
IP addresses remain unchanged.

Router Example

Organization Setup:

- Building A → Gigabit Ethernet LAN
- Building B → Gigabit Ethernet LAN
- LANs connected through high-speed backbone
- Router provides Internet connectivity

Figure 5.87 *Routing example*



Comparison of Connecting Devices

Device	Layer	Address Used	Filtering
Repeater/Hub	Physical	None	No
Switch	Data Link	MAC Address	Yes
Router	Network	IP Address	Yes

Summary

- Repeaters and hubs operate at the Physical Layer.
- Hubs broadcast frames to all ports.
- Switches operate at the Data-Link Layer.
- Switches use MAC addresses and filtering tables.
- Transparent switches learn MAC addresses automatically.
- Routers operate at the Network Layer.
- Routers connect different networks using IP addresses.
- Routers modify MAC addresses while forwarding packets.

Key Point

Hub = Broadcast Everything

Switch = Forward by MAC Address

Router = Forward by IP Address